

MECHANISMS CHEAT SHEET

DOF → GRASHOF → POSITION → VELOCITY →

CAMS → GEARS → DYNAMICS

1. DEGREE OF FREEDOM (Gr  bler)

Gr  bler’s equation:

M = 3(L - 1) - 2J1 - J2

- L = number of links (incl. ground)
- J1 = full joints (pin, slider)    1 DOF each
- J2 = half joints (higher pairs)    2 DOF each

Interpretation

M ≥ 1	Mechanism
M = 0	Structure
M < 0	Preloaded structure

Typical fourbar: L = 4, J1 = 4, J2 = 0 ⇒ M = 1

2. GRASHOF CONDITION

Let S = shortest, L = longest, P, Q = others.

S + L ≤ P + Q ⇒ Grashof

Mechanism types (Grashof)

Shortest link	Type
Ground (1)	Double-crank
Crank (2)	Crank-rocker
Coupler (3)	Double-rocker
Rocker (4)	Rocker-crank

Non-Grashof (S + L > P + Q) ⇒ **Rocker-rocker**
Equality (S + L = P + Q) ⇒ **Toggle / change-point**

3. FOURBAR POSITION ANALYSIS

Link labelling

a = crank, b = coupler, c = rocker, d = ground

Vector loop equations

→R2 + →R3 - →R4 - →R1 = →0

X: a cos θ2 + b cos θ3 - c cos θ4 - d = 0

Y: a sin θ2 + b sin θ3 - c sin θ4 = 0

Known: a, b, c, d, θ2. Solve for: θ3, θ4.

Freudenstein constants

K1 = d/a K2 = d/c K3 = (a²-b²+c²+d²)/2ac

K4 = d/b K5 = (c²-d²-a²-b²)/2ab

Coefficients for θ4

A = cos θ2 - K1 - K2 cos θ2 + K3

B = -2 sin θ2

C = K1 - (K2 + 1) cos θ2 + K3

θ4 = 2 arctan ((-B ± √(B² - 4AC)) / 2A)

Coefficients for θ3

D = cos θ2 - K1 + K4 cos θ2 + K5

E = -2 sin θ2

F = K1 + (K4 - 1) cos θ2 + K5

θ3 = 2 arctan ((-E ± √(E² - 4DF)) / 2D)

+ branch ⇒ **open** configuration

- branch ⇒ **crossed** configuration

FOURBAR.P outputs both automatically.

4. FOURBAR VELOCITY ANALYSIS

Differentiated loop equations

X: -aω2 sin θ2 - bω3 sin θ3 + cω4 sin θ4 = 0

Y: aω2 cos θ2 + bω3 cos θ3 - cω4 cos θ4 = 0

Known: θ2, θ3, θ4, ω2. Solve for: ω3, ω4.

Cramer’s rule solution

ω3 = (a ω2 sin(θ4 - θ2) / b sin(θ3 - θ4)) ω4 = (a ω2 sin(θ3 - θ2) / c sin(θ3 - θ4))

Sign: +ω = CCW, -ω = CW.

FOURBAR.V requires θ3, θ4 from position step.

5. CRANK-SLIDER POSITION

Variables

a = crank, b = coupler, c = offset, d = slider displacement

Loop equations

a cos θ2 - b cos θ3 - d = 0

a sin θ2 - b sin θ3 - c = 0

Solution

θ3 = arcsin ((a sin θ2 - c) / b)

d = a cos θ2 - b cos θ3

Zero offset: c = 0 (in-line crank-slider).

6. CRANK-SLIDER VELOCITY

ω3 = (a ω2 cos θ2 / b cos θ3)

ḃ = -a ω2 sin θ2 - b ω3 sin θ3

ḃ is the linear velocity of the slider.

7. TRANSMISSION ANGLE

Angle μ between coupler (b) and output link (c).

cos μ = (b² + c² - a² - d² + 2ad cos θ2) / 2bc

μ ≈ 90°	Excellent
μ > 40°	Acceptable
μ < 40°	Poor �� redesign

TRANS_ANG outputs rating automatically.

8. KEY DEFINITIONS

- **Mechanism**    transmits/transforms motion
- **Machine**    mechanism + significant force
- **Kinematics**    motion without forces
- **Kinetics**    forces causing motion
- **Binary/Ternary/Quaternary**    2/3/4 joints on a link

- **Full joint** (1 DOF): pin, slider
- **Half joint** (2 DOF): roll-slide, cam
- **Kinematic chain** “links + joints $\Rightarrow$ controlled motion”

9. EXAM STRATEGY

1. Compute DOF “state result (mechanism / structure)
2. Check Grashof “identify mechanism type
3. Position analysis “write loop equations first, then solve
4. Velocity analysis “differentiate, use Cramer’s rule
5. Transmission angle “check quality if asked

ALWAYS: Write equations before plugging numbers.
ALWAYS: State which configuration (open/crossed).
ALWAYS: Include units (rad, rad/s, mm, etc.).
CALCULATOR: MAIN $\rightarrow$ select program $\rightarrow$ enter values.

10. VARIABLE QUICK REFERENCE

Symbol	Meaning	Unit
a	Crank length	mm
b	Coupler length	mm
c	Rocker / offset	mm
d	Ground / slider disp	mm
θ_2	Crank angle (input)	deg
θ_3	Coupler angle	deg
θ_4	Output/rocker angle	deg
ω_2	Input angular vel.	rad/s
ω_3	Coupler angular vel.	rad/s
ω_4	Output angular vel.	rad/s
M	Degrees of freedom	“
μ	Transmission angle	deg

11. CAM DESIGN

Notes: Cams/CamDesign.pdf Secs. 2–8.

C-F01 Law of cam design (Cam Sec. 4)
 s, v, a continuous; j finite. Dwell $\Rightarrow v = a = 0$ at boundary.

C-F02 3-4-5 polynomial (Cam Sec. 5)
 $s = h[10(\theta/\beta)^3 - 15(\theta/\beta)^4 + 6(\theta/\beta)^5]$

C-F03 Cycloidal (Cam Sec. 5)
 $s = h[\theta/\beta - \sin(2\pi\theta/\beta)/(2\pi)]$; $C_a = 2\pi$.

C-F04 SCCA family (Cam Sec. 5)

Mod. Trap: $C_a = 4.888$ (min. accel).
Mod. Sine: $C_v = 1.760$ (min. vel).

C-F05 Pressure angle (Cam Sec. 8)
 $\tan \phi = \frac{v - \varepsilon \omega}{(s + R_p)\omega}$; keep $\phi < 30^\circ\text{--}40^\circ$.

C-F06 Roller follower (Cam Sec. 8)
Roller radius $<$ min. pitch-curve radius of curvature $\rho_{\min}$.

12. GEARS & GEAR TRAINS

Notes: Gears/GearsNotes.pdf Secs. 1–9.

G-F01 Pitch & size (Gears Sec. 4)

$$p_c = \frac{\pi d}{N} = \pi m, \quad p_b = p_c \cos \phi, \quad P_d = \frac{N}{d}, \quad d = \frac{N}{P_d} = mN$$

$mP_d = 25.4$ (mm-in); AGMA default = inch.

G-F02 Velocity / torque (Gears Sec. 4)

$$m_V = \frac{\omega_o}{\omega_i} = \pm \frac{N_i}{N_o}, \quad m_T = 1/m_V, \quad m_G = \max(|m_V|, |m_T|) \geq 1$$

Ext. $-$; int. $+$.

G-F03 Tooth dims (AGMA) (Gears Sec. 4)

$$a = \frac{1}{P_d}, \quad b = \frac{1.25}{P_d}, \quad h_t = 2.25/P_d, \quad c = 0.25/P_d, \quad D_o = d + 2a$$

G-F04 Centre distance (Gears Sec. 4)
 $C = r_p + r_g$ (ext.); $C = r_g - r_p$ (int.).

G-F05 Contact ratio (Gears Sec. 5)

$$Z = \sqrt{(r_p + a_p)^2 - (r_p \cos \phi)^2} + \sqrt{(r_g + a_g)^2 - (r_g \cos \phi)^2} - C \sin \phi$$
$$m_p = Z/p_b = ZP_d/(\pi \cos \phi) \quad \text{require } m_p \geq 1.2$$

G-F06 Backlash (Gears Sec. 6)
 $r_b = r_p \cos \phi$ (fixed); $\phi_{\text{new}} = \cos^{-1}(r_b/r_{p,\text{new}})$

$$\theta_B = \frac{43200 \Delta C \tan \phi}{\pi d} \text{ [min arc]}$$

G-F07 Min pinion teeth (Gears Sec. 7)
 20° full-depth: ≥ 21 (hob), ≥ 18 (rack).

G-F08 Simple train (Gears Sec. 8)
 $m_V = \omega_i/\omega_o = \pm N_i/N_o$; idlers flip sign only.

G-F09 Compound train (Gears Sec. 8)
 $m_V = \pm (\prod N_{\text{drv}})/(\prod N_{\text{drn}})$.

G-F10 Reverted (Gears Sec. 8)
 $N_2 + N_3 = N_4 + N_5$ (equal centre distances).

G-F11 Planetary (2-step) (Gears Sec. 8)

$$(1) \omega_{\text{rel}} = \omega_{\text{abs}} - \omega_{\text{arm}}$$
$$(2) \frac{\omega_L - \omega_{\text{arm}}}{\omega_F - \omega_{\text{arm}}} = \pm \frac{N_F}{N_L}$$

Sun fixed: $\omega_p = (N_s/N_p + 1)\omega_{\text{arm}}$; ring fixed: $\omega_p = (1 - N_r/N_p)\omega_{\text{arm}}$.

G-F12 Pitch point law (Gears Sec. 2)
 $\omega_3/\omega_2 = O_2K/O_3K$; constant ratio $\Leftrightarrow$ pitch point fixed.

13. MACHINERY DYNAMICS

Notes: Dynamics/DynamicsNotes.pdf Secs. 2–11.

D-F01 Parallel axis (Dyn. Sec. 2)

$$I_{ZZ} = I_{GG} + md^2$$
; $k = \sqrt{I/m}$; $k > d$ always.

D-F02 Composite CG (Dyn. Sec. 2)
 $\bar{X} = \sum W_i x_i / \sum W_i$; cylinder $I_{GG} = \frac{m(3r^2 + l^2)}{12}$.

D-F03 Energy (Dyn. Sec. 7)
 $KE = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$; $PE = mgh + \frac{1}{2}kx^2$.

D-F04 D’Alembert (Dyn. Sec. 8)
 $\sum \mathbf{F} - m\mathbf{a} = \mathbf{0}$; $\sum T - I\alpha = 0$.

D-F05 Virtual work (Dyn. Sec. 9)
 $\sum F_k v_k + \sum T_k \omega_k - \sum m_k a_k v_k - \sum I_k \alpha_k \omega_k = 0$

D-F06 Effective load (Dyn. Sec. 10)
 $m_{\text{eff}} = (b/a)^2 m_B$; $k_{\text{eff}} = (a/b)^2 k_A$.

D-F07 Reflected inertia (Dyn. Sec. 10)
 $I_{\text{eff}} = I_{\text{load}}(\omega_{\text{load}}/\omega_{\text{ref}})^2$

D-F08 IPS default (Dyn. Sec. 11)
 $g = 386 \text{ in/s}^2$; mass in blob: $m = W/g$.

14. GLOSSARY (selected)

Term	Meaning
Pitch point P	Fixed point on line of centres
Contact point Q	Moving tooth contact
Blob (bl)	IPS mass; $m = W/386$
m_p	Contact ratio; ≥ 1.2
Arm	Planetary carrier
SVAJ	Cam s, v, a, j program
μ	Transmission angle
Grashof	$S + L \leq P + Q$

15. FORMULA INDEX

Linkage (Secs. 1–10): on this sheet only
Combined: Weekly Notes/MechanismsStudyNotes.pdf

C-F01–F06	Cam Design notes Secs. 4–8
G-F01–F12	Gears notes Secs. 2–8
D-F01–F08	Dynamics notes Secs. 2–11
Glossary	MechanismsStudyNotes.pdf (appendix)